

SRM VEC
DEPARTMENT OF AIDS
Semester End Examination
(Regulations 2023)
AD3445 – AI1

Branch: B.Tech | Date: 01.02.2026 | Max. Marks: 100 | Duration: 3 Hrs

Answer ALL Questions
(100 Marks)

PART A — (10 × 2 = 20 marks)

1. Define a Rational Agent and list the four factors that depend on the measure of rationality. [CO1] [BTL1-Remember] (2)
2. Differentiate between informed and uninformed search strategies with an example for each. [CO1] [BTL2-Understand] (2)
3. What is a Constraint Satisfaction Problem (CSP)? Mention its three components. [CO2] [BTL1-Remember] (2)
4. Explain the purpose of Alpha-Beta pruning in game tree search. [CO2] [BTL2-Understand] (2)
5. Convert the following sentence into First Order Logic (FOL): 'Every student who takes AI loves programming.' [CO3] [BTL3-Apply] (2)
6. Define Skolemization in the context of converting FOL to Clausal Form. [CO3] [BTL1-Remember] (2)
7. State Bayes' Rule and explain the significance of the 'Likelihood' term. [CO4] [BTL2-Understand] (2)
8. What is a Markov Blanket in a Bayesian Network? [CO4] [BTL1-Remember] (2)
9. Distinguish between Partial Order Planning and Total Order Planning. [CO5] [BTL2-Understand] (2)
10. Define Information Gain in the context of Decision Tree Learning. [CO5] [BTL1-Remember] (2)

PART B — (5 × 16 = 80 marks)

11(a) (i) Explain the A* Search algorithm. Prove that A* is optimal if the heuristic $h(n)$ is admissible. (ii) Compare Breadth-First Search and Depth-First Search based on time complexity, space complexity, and completeness.

OR

(b) (i) Describe the PEAS description for an Automated Taxi Driver system. (ii) Explain the Hill Climbing search algorithm and discuss the problems of Local Maxima, Plateaus, and Ridges.

12(a) (i) Illustrate the Minimax algorithm with a sample game tree of depth 3 and branching factor 2. (ii) Solve the following Cryptarithmic problem: SEND + MORE = MONEY. Show the steps and constraints.

OR

(b) (i) Explain the backtracking search for CSPs and discuss how the 'Minimum Remaining Values' (MRV) and 'Degree Heuristic' improve its efficiency. (ii) Describe the concept of Arc Consistency (AC-3 algorithm) in CSPs.

13(a) (i) Explain the Resolution procedure in Propositional Logic with an example. (ii) Describe the steps involved in converting a First-Order Logic formula into Conjunctive Normal Form (CNF).

OR

(b) (i) Compare Forward Chaining and Backward Chaining inference techniques in First-Order Logic. (ii) Represent the following knowledge using Semantic Networks: 'Mammals are animals. Dogs are mammals. Tommy is a dog. Dogs have tails.'

14(a) (i) Consider a Bayesian Network with three nodes: Rain (R), Sprinkler (S), and Grass Wet (W). Given $P(R)=0.2$, $P(S|R)=0.01$, $P(S|\sim R)=0.4$, $P(W|S,R)=0.99$, $P(W|S,\sim R)=0.9$, $P(W|\sim S,R)=0.8$, $P(W|\sim S,\sim R)=0.0$. Calculate $P(R|W)$. (ii) Explain the components and semantics of a Bayesian Network.

OR

(b) (i) Explain the Dempster-Shafer theory for handling uncertainty and compare it with Bayesian reasoning. (ii) Describe the Variable Elimination algorithm for exact inference in Bayesian Networks.

15(a) (i) Explain the ID3 algorithm for Decision Tree induction. Illustrate with a small dataset how Entropy and Information Gain are calculated. (ii) Describe the components of a planning problem in PDDL (Planning Domain Definition Language).

OR

(b) (i) Explain the GraphPlan algorithm and how it uses a Planning Graph to find a solution. (ii) Discuss the different types of Machine Learning: Supervised, Unsupervised, and Reinforcement Learning.